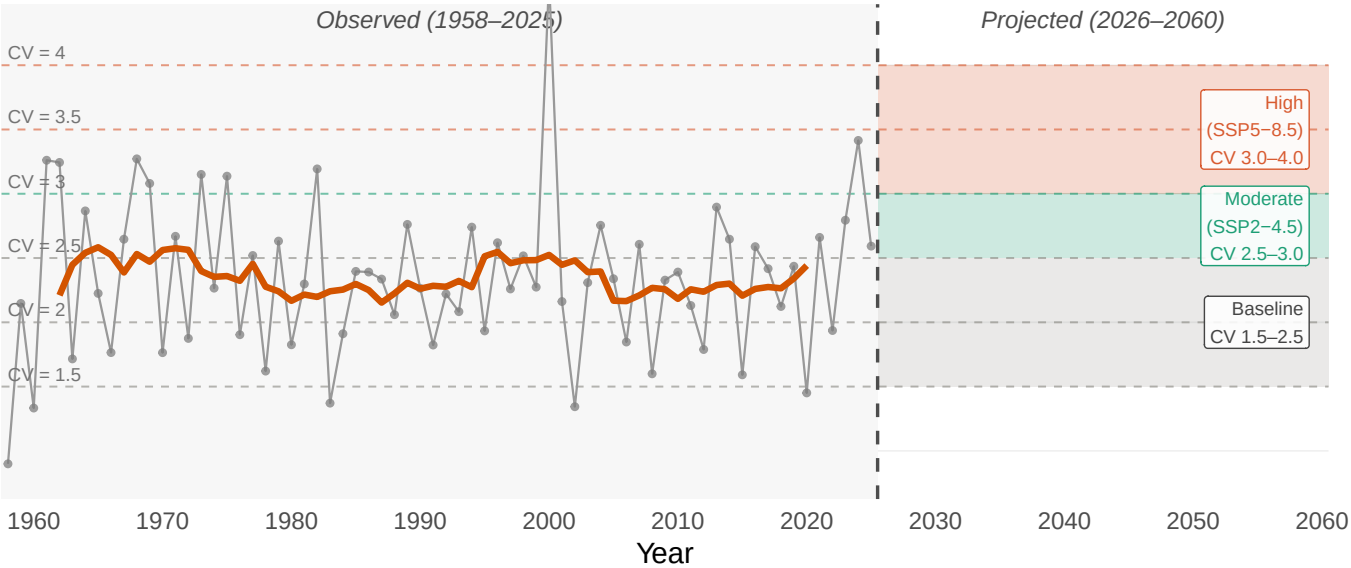


Historical CV and Projected Future Variability — Lishan Station

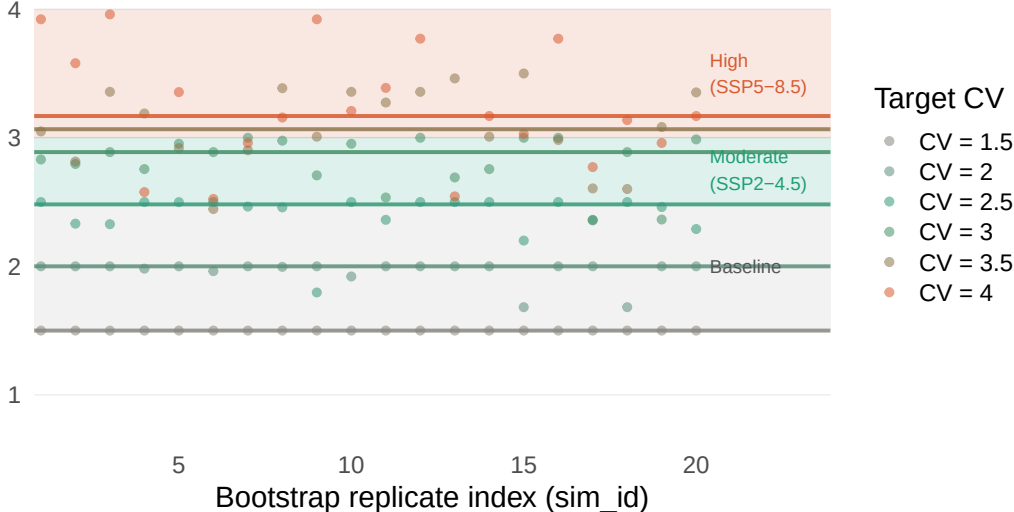
Grey points = annual CV; orange = 10-yr rolling mean; dashed lines = FFF target CV levels. Projected bands are literature-based scenario proxies (not direct GCM output).



Observed: Lishan Station 01T230 (WRA), 1958–2025. Projected CV ranges are scenario proxies for eastern Taiwan mountain streams: Baseline CV 1.5–2.5 (this study); Moderate variability CV 2.5–3.0 (SSP2–4.5 proxy: Shiau & Huang 2014, J. Hydro–environ. Res. 8(4):355–366; Tung et al. 2016, Paddy Water Environ. 14(1):15–26); High variability CV 3.0–4.0 (SSP5–8.5 proxy: Tung et al. 2016; IPCC AR6 WGI Ch.11, 2021). CV ranges are sensitivity bounds, not deterministic GCM projections.

Option B: Bootstrap ensemble all simulations by scenario

Each point = one bootstrap year; horizontal line = median per CV level. Colour = target CV level (scenario intensity).



n = 120 total simulations across 6 CV levels.

Left: Observed CV, Lishan Station 01T230 (WRA) 1958–2025. Right: Two-layer bootstrap ensemble (Layer 1 = block resample; Layer 2 = CV scaling, Arnell 1998). Scenario proxies: Shiau & Huang (2014); Tung et al. (2016); IPCC AR6 WGI Ch.11 (2021).